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On the Incoherencies in Web Browser Access Control Policies

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Abstract—Web browsers' access control policies have evolved Inconsistent principal labeling. Today's browsers do piecemeal in an ad-hoc fashion with the introduction of new not have the same principal definition for all browser re- browser features. This has resulted in numerous incoherencies. sources (which include the Document Object Model (DOM), In this paper, we analyze three major access control flaws in today's browsers: (1) principal labeling is different for different network, cookies, other persistent state, and display). For resources, raising problems when resources interplay, (2) run- example, for the DOM (memory) resource, a principal is time changes to principal identities are handled inconsistently, labeled by the origin defined in the same origin policy and (3) browsers mismanage resources belonging to the user (SOP) in the form of <protocol, domain, port> [4]; but principal. We show that such mishandling of principals leads for the cookie resource, a principal is labeled by <domain, to many access control incoherencies, presenting hurdles for web developers to construct secure web applications. path>. Different principal definitions for two resources are A unique contribution of this paper is to identify the com- benign as long as the two resources do not interplay with patibility cost of removing these unsafe browser features. To do each other. However, when they do, incoherencies arise. For this,

we have built WebAnalyzer, a crawler-based framework example, when cookies became accessible through DOM's for measuring real-world usage of browser features, and used "document" object, DOM's access control policy, namely the it to study the top 100,000 popular web sites ranked by Alexa. Our methodology and results serve as a guideline for browser SOP, undermines some of cookie's access control policies designers to balance security and backward compatibility. (Section II-C1 gives a more detailed analysis). Inappropriate handling of principal label changes. A web application is allowed to change its principal's label I. I NTRODUCTION at runtime through the use of the document.domain Web browsers have gradually evolved from an application DOM property. Nevertheless, the access control state is that views static web pages to a rich application platform often kept static and such "effective" principal IDs set by on which mutually distrusting web site principals co-exist document.domain are disregarded. This leads to access and interact [1]–[3]. Along the way, the browsers' access control incoherencies. control policies have also been evolving, but unfortunately Disregard of the user principal. In this paper, we this happened in a piecemeal and ad-hoc fashion alongside introduce the concept of a user principal in the browser the introduction of new browser features (such as AJAX) setting. The user principal represents the user of a browser. or resources (such as local storage). There have been no Sometimes, the user principal is disregarded in existing principles or invariants that a new access control policy must browsers' access control policies. Certain resources should follow or maintain. Consequently, numerous incoherencies belong to the user principal exclusively. They include the in browsers' access control policies exist, presenting hurdles user-private state such as clipboard and geolocation, user for web programmers to build robust web applications. actions like navigating back and forward in browsing history, In this paper, we examine the current state of browser ac- and a browser's UI including the current tab. These resources cess control policies, uncover and analyze the incoherencies should not be accessible by web applications without user in these policies, and measure the cost of eliminating them permission; otherwise, a web site could impersonate the user in today's web. and violate user privacy. Unfortunately, today's DOM APIs An access control policy configures how a principal expose some of these resources to web applications. accesses certain resources. This involves defining how prin- To systematically analyze and uncover the incoherencies cipals are identified, how resources are labeled with principal created by these three problem areas, we have devised a set IDs, and how these labels may be changed and handled at of coherency principles and constructed tests to check major runtime. Unfortunately, browsers often mismanage princi- browsers (including Internet Explorer, Firefox, and Google pals, resulting in access control inconsistencies. We focus on Chrome) for violations of these principles and to uncover three major sources of these problems: inconsistent principal the incoherencies that ensue. labeling, inappropriate handling of principal label changes, A major goal of our work is to evaluate the compatibility and disregard of the user principal. cost of removing unsafe browser features that contribute to the incoherencies. To this end, we have built WebAnalyzer, a 1) Each shared browser resource, i.e. a resource shared scalable, crawler-based browser-feature measurement frame- among multiple principals, should have a principal work that can inspect a large number of web pages by definition (labeling of principals that share the re-rendering them in instrumented browsers. WebAnalyzer cap- source) and have an access control policy. tures the DOM interactions of a page by interposing between 2) For each non-shared browser resource that is explicitly the JavaScript engine and the DOM renderer, captures the owned by a single principal, the resource should have protocol-level behavior through an HTTP

proxy, and an owner principal with a specific label or be globally analyzes the visual appearance of a page by extracting its page accessible layout. 3) When two resources interplay, both resources should be armed with WebAnalyzer, we have conducted measurements on the prevalence of unsafe browser features over the This is because when two resources have different most popular 100,000 web sites as ranked by Alexa [5]. Our ways of labeling principals and when they interplay, results pinpoint some unsafe features that have little back- their respective access control policies can be in con- ward compatibility cost and are thus possible to remove from current browsers without breaking many sites. For example, 4) All access control policies must consider the runtime we find that most APIs controlling user-owned resources, label of the principals, namely, the “effective” principal ID. quest’s principal labeling have low compatibility costs, 5) The user principal’s resources should not be accessible whereas a substantial number of sites depend on “dangerous” by web applications. functionality provided by document.domain or transparent- This is because when the user principal’s resources enter cross-origin overlapping frames. Overall, we believe that are accessible by web applications, the user’s privacy by estimating the prevalence of unsafe features on the web, may be compromised or a web application could act our measurements can guide future browsers to make better on the user’s behalf without the user’s knowledge. security vs. functionality trade-offs. We look for violations of these principles and check for In summary, this work makes the following contributions: incoherencies when violations take place. The pseudocode • A systematic, principal-driven analysis of access control incoherencies in today’s browsers. • Introduction of the user principal concept for the 0 foreach (browser resources) { browser setting. 1 if exists (access control) { • A comprehensive, extensible compatibility measure- 2 if !considers (effective principal ID) ment framework. 3 check improper principal ID changes • The first large-scale measurements on the compatibility 4 } else cost of coherent access control policies. 5 check if lack of policy is appropriate The rest of the paper is organized as follows. Section II 6 } presents our systematic analysis of today’s browser access 7 control policies and enumerates access control incoheren- 8 foreach (pairs of resources) { cies. Section III discusses our measurement motivation, 9 if (they interplay && tools, and infrastructure. Section IV presents our measure- 10 the principal/owner labeling differs) ment results and gives recommendations on which unsafe 11 check resource interplay incoherencies policies can be eliminated with acceptable compatibility 12 } cost. Section V discusses limitations of our approach, Section VI presents related work, and Section VII concludes. For each resource, we check whether it has an access II. A N ANALYSIS OF BROWSER ACCESS CONTROL control policy. If not, we check whether the lack of policy INCOHERENCIES is appropriate (line 5, for example, Section II-E illustrates on how some resources that belong to the user principal In this section, we present our systematic analysis of lack access control considerations). If yes, we further check today’s browser access control policies and enumerate their whether the access control policy considers the effective incoherencies. principal ID that sites can change dynamically at render- time. If it does not, then we check for incoherencies there A. Methodology (line 3, Section II-D). For a systematic analysis, we establish the following In addition, we go through all pairs of resources; if they access control coherency principles to guide our search for interplay and if they have the different principal definitions, incoherencies: we check for incoherencies (line 11, Section II-C). Careful Shared resources Principal definition documents using the Document Object Model (DOM), DOM objects SOP origin which is

the platform- and language-neutral interface that cookie domain/path allows scripts to dynamically access and update the content, localStorage SOP origin structure and style of a document [6]. sessionStorage SOP origin A cookie is a persistent state resource. The browser display SOP origin and dual ownership * ensures that a site can only set its own cookie and that Table I a cookie is attached only to HTTP requests to that site. SHARED BROWSER RESOURCES AND THEIR RESPECTIVE PRINCIPAL DEFINITIONS . DISPLAY ACCESS CONTROL IS NOT WELL- DEFINED IN TODAY ' S By default, the principal is labeled with the host name BROWSERS . and path, but without the protocol and the port number [7], [8], unlike SOP origins. For example, if the page a.com/dir/1.html creates a cookie, then that cookie Non-shared resources Owner is accessible to a.com/dir/2.html and other pages XMLHttpRequest SOP origin from that dir/ directory and its subdirectories, but is not postMessage SOP origin accessible to a.com/. Furthermore, https://a.com/ clipboard user and http://a.com/ share the cookie store unless a browser history user* cookie is marked with a "secure" flag. Non-HTTPS sites geolocation user can still set "secure" cookies in some implementations, but Table II cannot read them back [9]–[11]. A web programmer can NON - SHARED BROWSER RESOURCES AND THEIR RESPECTIVE OWNER PRINCIPAL . ACCESS CONTROL IS NOT WELL- DEFINED IN TODAY ' S BROWSERS . make cookie access less restrictive by setting a cookie's domain attribute to a postfix domain or the path name to be a prefix path. Local storage is the persistent client-side storage shared readers may wonder what happens to the interplay of more among principals defined by SOP origins [12]. than two resources. Coherency in this context is a transitive Session storage is storage for a tab [12]. Each tab has a property. That is, if a Resource 1 and Resource 2's access unique set of session storage areas, one for each SOP origin. control policies are coherent (namely have the same princi- The sessionStorage values are not shared between tabs. The pal definitions) and that of Resource 2 and Resource 3 are lifetime of this storage is the same as that of the tab. coherent, then the access control policies of Resource 1 and Display does not have a well-specified access control Resource 3 are also coherent since their principal definitions policy in today's browsers and standards (corresponding to should also be the same. line 5 in our pseudocode). Our earlier work Gazelle [3] The enumeration of resources is done by manually brows- specified an access control policy for display (and Gazelle ing through IE's source code (more in Section II-B). Our further advocated that this policy be enforced by the browser incoherency checks are done through test programs on major kernel, unlike existing browsers). In Gazelle's model, a web browser versions. site principal delegates its display area to another principal in Despite our effort to be comprehensive, it is possible that the form of cross-domain iframes (or objects, images). Such we miss some browser resources or miss some interplays an iframe (window) is co-owned by both the host page's among the resources. We hope our work to be a start for principal, called landlord, and the nested page's principal, a community effort on mapping out the full set of browser called tenant (both labeled with SOP origins). Principals access control policies. other than the landlord and the tenant have no access per- missions for the window. For the top-level window, the user B. Browser resources principal owns it and plays the role of its landlord. Gazelle's In this section, we enumerate all types of browser re- policy further specifies how landlord and tenant should sources. A browser resource may be shared among (some access the four attributes of a window, namely the position, definition of) principals or may not be shared and is ex- dimensions, pixels, and URL location. This specification plicitly owned by some principal. Table I shows the shared guarantees that the tenant cannot interfere with the landlord's resources and their respective principal definitions. Table II display, and that the tenant's pixels, DOM objects, and shows non-shared resources and their respective owners. navigation history are private to the tenant. Gazelle's

policy We now describe each resource, their principal or owner is coherent with SOP. In Table III, we summarized the access definition, and its access control policy in turn. control matrix for Gazelle, IE 8, Firefox 3.5, and Chrome 2. A DOM object is a memory resource shared among The access control of the URL location attribute corresponds principals labeled with SOP origins, namely, <protocol, to the navigation policy of a browser. Descendant navigation domain, port>. The access control policy of DOM objects is policy allows navigating a descendant window regardless governed by SOP [4], which mandates that two documents of its origin; this was advocated and implemented over from different origins cannot access each other's HTML several browsers [13]. Gazelle's policy is child navigation

Landlord Tenant Gazelle IE FF/Chrome Gazelle IE FF/Chrome position (x,y,z) RW RW RW RW dimensions (height, width) RW RW RW R RW R pixels W W* RW RW RW RW URL location W W RW* RW RW RW Table III

A CCESS CONTROL POLICY FOR A WINDOW ' S LANDLORD AND TENANT (BEING A DIFFERENT PRINCIPAL FROM THE LANDLORD) ON G AZELLE , IE 8, F IREFOX 3.5, AND C HROME . RW*: T HE URL IS READABLE ONLY IF THE LANDLORD SETS IT. IF THE TENANT NAVIGATES TO ANOTHER PAGE, LANDLORD WILL NOT SEE THE NEW URL. W*: THE LANDLORD CAN WRITE PIXELS WHEN THE TENANT IS TRANSPARENTLY OVERLAID ON THE LANDLORD .

policy. (We elaborate in Section II-C3 that the descendant navigation policy is at conflict with DOM's SOP.) Our tests indicate that Firefox 3.5 and Chrome 2 currently support the child policy, while IE 8 supports the descendant policy. All major browsers allow any window to navigate the top- level window, while Gazelle only allows top-level window navigation from the top-level window's tenant and the user. XMLHttpRequest allows a web site principal to use scripts to access its document origin's remote data store by issuing an asynchronous or synchronous HTTP request to the remote server [14]. XMLHttpRequest2 [15] and XDomainRequest have been recently proposed and implemented in major browsers to allow cross-origin communications with remote servers, where HTTP authentication data and cookies are not Figure 1. Incoherency arises from the interplay between the access control sent by default. These networking capabilities are not shared policies of DOM and cookies and strictly belongs to a web site principal labeled with a SOP origin. PostMessage is a recently proposed client-side cross- origin communication mechanism that is now implemented by setting the cookie with the "secure" flag. However, a in all major browsers. This is also a web site principal's "secure" cookie can still be set by an HTTP response capability which is not shared with any other principals. and be accessed by scripts belonging to an HTTP page as The last three resources in the non-shared resource table, long as their domains are the same. Additionally, different namely clipboard, browser history, and geolocation, all be- services running on different ports of the same domain can long to the user principal, and web applications should not be access each other's cookies. Moreover, the path protection able to access them directly. However, they are all accessible of cookies becomes ineffective as a script from a different by scripts through the DOM API, causing problems that we path can access the cookie based on SOP. describe in Section II-E. The interplay between DOM and cookies also allows the scripts to set the effective domain of a cookie to any suffix C. The interplay of the resources of the original domain by setting the domain attribute of From the enumeration of the resources and their respective the cookie. This can lead to inconsistencies in the current principal or owner definition in the above section, we derived browsers. Figure 1 shows a scenario in which such inconsis- the following problematic pairs of resources, where the two tencies lead to an undefined behavior in the browsers. In this resources interplay and their principal or owner definitions example, a cookie named "stockCookie" with value "buy" differ: DOM-cookie, cookie-

XMLHttpRequest, and DOM- is stored in the cookie store for the domain a.com. A script display. We elaborate on these interplays below. injected into a compromised page belonging to x.a.com

1. DOM and Cookies: DOM and cookies interplay be- can create another cookie with the same name but with a

cause scripts are able to create or modify cookies by using different value "sell" while setting its domain attribute to the document.cookie property in the DOM API. a.com. With no protocol in cookie's principal definition, cookies While this leads to a compromised state in the current are vulnerable to information leaks. A cookie intended for browsers, different browsers deviate in their behavior cre- a secure HTTPS principal can be passed over HTTP and ating further inconsistencies in the web applications sup- be exposed to network attackers. This can be prevented porting multiple browsers. Firefox 3 sets this cookie with a domain value of .a.com resulting in multiple cookies a window, a resource created by its descendant through a with the same name in browser's cookie store. The browser DOM API, even if the landlord and the descendant are attaches both cookies (genuine cookie with domain a.com different principals. This gives a malicious landlord more and evil cookie with domain .a.com) to any server requests powerful ways to manipulate a nested, legitimate sites than to a.com. The server only receives the cookie's name- just overdrawn: with overdrawn, a malicious landlord value pair without any information about its corresponding can imitate a tenant's content, but the landlord cannot send domain. This results in the server receiving two cookies with messages to the tenant's backend in the name of the tenant. the same name. Since server-side behavior is not defined in As an example attack, imagine that an attacker site nests case of duplicate cookies [9], it leads to inconsistent state at a legitimate trading site as its tenant. The trading site a.com's server. In case of IE 8, the original cookie value further nests an advisory site and uses a script to interact is overwritten and only the wrong cookie value is received with the advisory window to issue trades to the trading by the server. site backend (e.g., making a particular trade based on the

1. Cookies and XMLHttpRequest: Cookies and XML- advisory's recommendation shown in the URL fragment).

HttpRequest interplay because XMLHttpRequest can set With just one line of JavaScript, the attacker could navigate cookie values by manipulating HTTP headers through the advisory window (which is a descendant) and create scripts. XMLHttpRequest's owner principal is labeled by the unintended trades. SOP origin, while cookie has a different principal definition Another conflict lies in the access control on the pixels (Section II-B). of a window. DOM objects are ultimately rendered into If a server flags a cookie as "HttpOnly", the browser the pixels on the screen. SOP demands non-interference prevents any script from accessing (both reading and writing) between the DOM objects of different origins. However, the cookie using the document.cookie property. This existing browsers allow intermingling the landlord's and effectively prevents cookies being leaked to unintended tenant's pixels by overlaying transparent tenant iframes on parties via cross-site scripting attacks [16]. the landlord, deviating from the non-interference goal of The purpose of HttpOnly cookies is that such SOP. This enables an easy form of clickjacking attacks [19]. cookies should not be touched by client-side In contrast, Gazelle advocates cross-principal pixel isolation scripts. However, XMLHttpRequests are created and in accordance with SOP (Table III, row "pixels"). invoked by client-side JavaScript code, and certain methods of the XMLHttpRequest object facilitate D. Effective Principal ID access to cookies: getResponseHeader and Browsers allow

cross-principal sharing for “related” sites getAllResponseHeaders allow reading of the by allowing sites to change their principal ID via the “Set-cookie” header, and this header includes the value of document.domain property [4]. This property can be HttpOnly cookies. Another method, setRequestHeader, set to suffixes of a page’s domain to allow sharing of enables modification of this header to allow writing to pages across frames. For example, a page in one frame HttpOnly cookies. from x.a.com and a page from www.a.com initially Some of the latest browsers have tried to resolve this issue cannot communicate with each other due to SOP restrictions. with varied success. IE 8 currently prevents both read and This is one of the few methods for cross-origin frames to write to cookies via “Set-cookie” header, but still allows communicate before the advent of postMessage [20]. How- access via “Set-cookie2” header [17]. Firefox has also recog- ever, changing document.domain violates the principle nized and fixed the issue for cookie reads: their fix prevents of least privilege: once a subdomain sets its domain to its XMLHttpRequest from accessing cookie headers of any suffix, there is no control over which other subdomains can response, whether or not the HttpOnly flag was set for those access it. cookies [18]. This is a bold step taken by Firefox, as our Furthermore, almost no existing access control policies results show that a considerable number of web pages still of today’s browsers take such “effective” principal IDs into read cookie headers from XMLHttpRequest (Section IV). consideration. In the following subsections, we examine However, we have still observed the writing issue with how the disregard of effective principal IDs leads to dual HttpOnly cookies using Firefox 3.5. A script can set a cookie identities and incoherencies exploitable by attackers. In our with the same name as the HttpOnly cookie and can have a attack model, an attacker owns a subdomain (through third- different value set using the setRequestHeader method. party content hosting as in iGoogle or by exploiting a site This results in a duplicate cookie being sent to the server, vulnerability). As we will show in the following sections, thus creating an inconsistent state on the server side. the attacker can leverage document.domain to penetrate

1. DOM and Display: One incoherence takes place on the base domain and its other subdomains. URL location of a window. The descendant navigation 1) Cookie: Any change of origin using policy (Section II-B) is at conflict with DOM’s SOP. De- document.domain only modifies the effective principal scendant navigation policy allows a landlord to navigate ID for DOM access and does not impact the domain for Figure 2. Lack of effective principal ID consideration in cookie’s access Figure 3. Lack of effective principal ID consideration in XMLHttpRequest- control policy quest’s access control policy

cookie access. Figure 2 shows an attack to exploit this into 1.html via DOM access, and this script can then inconsistent behavior of browser policy design. In this make XMLHttpRequest calls to the original domain of the scenario, a page 1.html in domain x.a.com changes it page. Since a well-crafted XMLHttpRequest can change the effective domain to a.com. As a result, it can access the server-side state for the web application, and this state might DOM properties of other pages belonging to a.com, but be shared between other pages within the domain x.a.com, it can no longer access the pages of its original domain such attack can possibly impact all pages belonging to x.a.com. However, since the effective domain does not x.a.com. change for cookie access, the page still maintains access 3) postMessage: postMessage also ignores any to the cookies belonging to its original domain. This document.domain changes: if x.a.com changes do- inconsistent dual identity possessed by the page acts as a main to a.com and sends a message to y.b.com, bridge to

access cookies from both the original domain and y.b.com still sees the message's origin as x.a.com. Also, the effective domain. if y.b.com changes its domain to b.com, x.a.com still In order to launch the attack, an attacker (after owning has to address messages to y.b.com for them to be deliv- a subdomain page) first assumes the identity of a.com ered. This gives the attacker (with a compromised subdo- and subsequently injects a script into the page 1.html. main) an opportunity to send messages while masquerading This injected script can now read and write the cookies under the identity of another subdomain (Figure 4). belonging to x.a.com including any cookies created later. 4) Storage: Based on our tests, IE 8 does not take any Effectively, if the attacker can compromise a page in one document.domain changes into consideration for both of the subdomains, he can access the cookies of any other local storage and session storage. Firefox 3.5 also ignores subdomains that change their effective origin to the base effective principal ID for local storage. However, for session domain. storage, any domain changes via document.domain are

1. XMLHttpRequest: Change of origin for scripts does considered: the old session storage is lost for the original

not change the effective principal ID for XMLHttpRequest domain and a new session storage is created for the effective usage. This enables a (malicious) script in a (compromised) principal. subdomain to issue XMLHttpRequest to the servers be- Inconsistency arises when document.domain changes longing to the base domain and its other subdomains. The are ignored (for both session storage and local storage in IE; attack scenario is illustrated in Figure 3. Page 1.html for only local storage in Firefox). An attacker (being able to has changed its effective domain value to a.com from the inject a script into one of the pages of any subdomain, say original value of x.a.com. With no effect on XMLHttpRequest- x.a.com) can change its origin to the base domain a.com quest usage, scripts in 1.html can still make requests and can successfully inject a script into the DOM of the base to the server belonging to x.a.com. This again gives a domain or any other origins (e.g., y.a.com) that change script a dual identity – one for DOM access (a.com) and identity to the base domain. Since access control checks another for XMLHttpRequest (x.a.com). Therefore, an on storage rely on original domain (i.e., y.a.com), the attacker compromising any subdomain can inject a script malicious script can now freely access the storage belonging The window object has a history property with an array of user-visited URLs. Browsers have been denying any site's access to this array to protect user privacy, but they do allow a site to navigate the browser back and forward in history through the back() and forward() methods [8]. Worse, our tests indicate that Firefox 3 and Google Chrome 2 allow any child window to navigate the top-level window back or forward in history irrespective of the origin. In many cases this is just a nuisance, but some properly-crafted history navigation by a malicious application can lead to more severe damage. For example, the user might be tricked to make multiple purchases of the same product. We have also investigated synthetic event creation. The DOM API allows a site to generate synthetic mouse or key- board events through the document.createEvent() method (or document.createEventObject() in IE). Figure 4. Lack of effective principal ID consideration in postMessage In IE, a programmer could directly invoke a click() method on any HTML element to simulate user clicks. These techniques are useful for debugging purposes. To our delight, all major browsers are careful not to let a web site to manip- to y.a.com. ulate another site's user experience with these synthetic user events. Note that it is benign for a site to simulate the user's E. The User Principal actions for itself, since loading and rendering site content can by itself achieve any effects

of simulating user actions In this paper, we introduce the concept of the user principal (e.g., simulating a mouse click is equivalent of calling the `click` in the browser setting. The user principal represents the user of the browser. Unfortunately, it has often been neglected in browser access control policies. is the current tab window, or top-level window. In today's While a web application does manage the user's data browsers, any web site loaded in any window is able to reposition and resize a top-level window through the `moveTo`, `banking` data at a banking site), certain browser resources `moveBy`, `resizeTo`, and `resizeBy` properties of the window or data belong to the user exclusively and should not be accessible by any web site without user permissions. Such window effectively resizes the browser window. Firefox 3 resources include: user's private data, such as clipboard allows an application to resize a browser window even in the data and geolocation; user actions, such as clicking on presence of multiple tabs, while IE 8 and Chrome 2 do not the forward and back button; devices, such as camera and allow this. A site can also open and close a top-level window microphone; and browser UI, including the current tab using `open` and `close` methods. The use of `open` method window (top-level window). has been mitigated through built-in popup blockers. IE 8 Unfortunately, in today's browsers, some of these re- allows any frame to close a top-level window irrespective of sources are directly exposed to web applications through the origin, while Firefox 3 and Chrome 2 prevent this from the DOM API. This breaks the fundamental rule of protection: something bad is happening. These capabilities allow an attacker site (even detecting resources belonging to different principals from one when deeply nested in the DOM hierarchy, say a malicious another, as the user principal's resources can be accessed and) to directly interfere with the user's experience with the browser and manipulated by site principals. This can result in privacy compromises, information leaks, and attacks that trick Some of the other loopholes in browser UI have already users into performing unintended actions. In this section, been fixed. For example, the status bar can no longer be set we examine the user principal resources and describe our findings on how they may be accessed improperly by web 3) User-private state: Jackson et al. have shown that a applications. user's browsing history can be exposed by inspecting the

1. User actions: The focus and blur properties of the color of a visited hyperlink [21], raising privacy concerns.

window object allow web sites to change focus between the The hyperlink's color is intended for the user, and it is not windows that they opened irrespective of the origins. This necessary for web sites to be able to read it. enables an attacker site to steal focus or cause the user to The clipboard data also belongs exclusively to the act on a window unintentionally. user principal. All versions of IE since 5.0 support APIs to access clipboard data. A web site can get III. THE WEBAANALYZER MEASUREMENT FRAMEWORK contents of a user's clipboard by successfully calling `window.clipboardData.getData("Text")`. To achieve consistent browser access control policies, browser vendors need to remove or modify the browser may prompt user before getting the data. However, features that contribute to incoherencies. For example, the prompt does not identify the principal making the request, disallowing domain-setting for cookies, eliminating request (simply using the term "this site"). As a result, a document.domain, and removing support for access- malicious script embedded on a third-party frame may trick ing user principal resources are steps towards secure new the user into giving

away his clipboard because he thinks browsers. However, this begs the question of what the cost of that such access is being requested by the trusted top-level these feature removals is and how many web sites will break site. as a result. In today's highly competitive browser market, backward compatibility with the existing web is paramount. Geolocation is one of the latest browser features that To help browser vendors balance security and compati- allows a site to determine the client's location by using the bility, we set off to build a measurement system to measure navigator.geolocation [12] interface. At the time the cost of security. Many previous web compatibility studies of writing, Firefox 3.5 is the only stable production browser have been browser-centric: they have evaluated the degree to supporting this HTML5 feature. Geolocation is user-private which a given browser supports various web standards or is data. Today's browsers do ask user permission before access- vulnerable to attacks [22], [23]. In contrast, we take a web- ing it. However, issues arise when a site embeds content centric perspective and actively crawl the web to look for from multiple principals (i.e., in frames), and more than prevalence of unsafe browser features on existing web pages. one origin needs access to geolocation information. The Compared to existing crawlers, however, static web page geolocation dialog is active for only one origin at a time; inspection is insufficient. Dynamic features such as AJAX if there is a request to access geolocation from b.com or post-render script events require us to actively render a while the dialog for a.com is still active, it is ignored web page to analyze its behavior at run time. Moreover, the — the principal that succeeds in invoking the geolocation incoherencies we identified in Section II require analysis of dialog first wins. Therefore, if a malicious party manages to not just a page's JavaScript execution [24], but also DOM embed a script (or a frame) on the page, it can prevent the interactions, display layout, and protocol-layer data. main site from triggering the geolocation dialog by invoking the dialog first. As a result, the malicious party can create To address these challenges, we have constructed a scal- denial-of-service against the main site, preventing it from able, execution-based crawling platform, called WebAna- retrieving a user's geolocation information. Additionally, it lyzer, that can inspect a large number of web pages by could trick the user into giving away location to itself rather rendering them in an instrumented browser. The platform than the main site (e.g., using phishing domain names like consumes a list of URLs (defined by a human operator or www.google.com). generated by a traditional web crawler), and distributes them among virtual machine workers, which renders them using Changing document.domain also generates inconsis- IEWA , a specially instrumented version of Internet Explorer. tencies. The geolocation prompt is designed to work only IEWA provides dynamic mediation for all browser resources, with the original principals, and even if a site changes and detects when a resource invocation matches one of identity, the prompt still displays the original domain as preset policy rules. Even though our framework is extensible the requesting domain. For an example site good.a.com to a large variety of browser policies, we concentrate on “un- that changes its document.domain to a.com, this causes the safe feature” rules derived from our analysis in Section II. following problems: To build IEWA , the central piece of our measurement • If an attacker site evil.a.com changes its platform, we leverage public COM interfaces and exten- document.domain to a.com, it can steal position sibility APIs exported by Internet Explorer 8. Figure 5 information from good.a.com, if good.a.com shows the architecture of IEWA , which centers around three has stored or displayed this information in major interposition modules: (1) a script engine proxy, which a place that is accessible via the DOM (e.g., provides JavaScript and DOM interposition, (2) a network using parent.document.getElementById(

proxy based on Fiddler [25], and (3) display dumper, which "coords").innerHTML). enables custom analysis of a page's layout as it is visible to • If another site evil.a.com also changes its domain the user. Next, we discuss each module in turn. to a.com, it could impersonate good.a.com, Script engine proxy. We build on our earlier system by using parent.navigator.geolocation in MashupOS [1] to implement a JavaScript engine proxy .getCurrentPosition, which would trigger (called script engine proxy (SEP)): SEP is installed be- the access prompt using good.a.com, instead of tween IE's rendering and script engines, and it mediates evil.a.com. and customizes DOM object interactions. SEP exports the Figure 5. High-Level Architecture of IEWA .

script engine API to IE's renderer, and it exports the DOM consecutive URLs, detects when sites fail to render (e.g., and rendering interfaces to IE's script engine. Each DOM 404 errors), and recovers from any browser crashes. object is interposed by a corresponding object wrapper. Visiting a site's home page is sometimes insufficient to When IE's script engine asks for a DOM object from the invoke the site's core functionality. For example, a feature rendering engine, SEP intercepts the request, retrieves the may be accessed only when the user clicks on a link, types corresponding DOM object, associates the DOM object with search queries, or causes mouse event handlers to run. its wrapper object inside SEP, and then passes the wrapper It is difficult and time-consuming to fully automate a site's object back to the original script engine. Any subsequent analysis to study all possible features and pages that could invocation of wrapper object methods from the original be invoked using all combinations of user input. Instead of script engine passes through SEP. SEP is implemented as aiming for complete coverage within a particular site, we a COM object and is installed into IE by modifying IE's enhanced our navigation engine with simple heuristics that JavaScript engine ID in the Windows registry. simulate some user interaction. After rendering a site's home Network interposition. In addition to SEP, we route the page, IEWA will find and simulate a click on at most five browser's network traffic through a proxy to monitor all random links, producing five random navigation events. In HTTP/HTTPS requests and analyze cookie transfers as well addition, IEWA will check for presence of a search form, as network APIs like XMLHttpRequest. Our network proxy fill it with random keywords, and submit it. We restrict all is implemented using the FiddlerCore interfaces provided by simulated navigations to stay within the same origin as a the public-domain Fiddler web debugging proxy [25], [26]. site's home page. Display analysis. In order to evaluate display policies, it These simple enhancements maintain our ability to ex- is necessary to analyze a browser's visual output as seen by amine a large number of sites while adding the ability to the user. For this purpose, we use a customized version of properly handle many (but not all) sites with home pages IE's rendering engine that exposes COM interfaces to extract that do not invoke the site's main functionality. For example, a textual representation of a particular page's visual layout we can navigate to a random article on Wikipedia, a random at any stage of rendering. In our current evaluation, we use video on YouTube, a random profile on MySpace, a random these COM interfaces to save a snapshot log of IE's display Twitter feed, and a random search query on Google. We after a page has fully loaded. Because some pages have evaluate the success of this methodology against a user- post-render events that alter layout, we wait an additional driven browsing study in Section IV-G and discuss its 5 seconds before taking a display snapshot. Snapshot logs limitations in Section V. provide a mapping between a page's objects and their layout Performance. We deployed our system on several desktop properties, such as position, dimensions, or transparency. machines, each with an Intel 2.4 GHz quad-core CPU and They can be analyzed offline for the presence of unsafe 4 GB of RAM. Our IEWA workers

run inside a Windows frame overlapping behavior or other dangerous page layouts. Vista VMware virtual machine to prevent malware infection. Navigation. To facilitate automatic analysis for a large We executed multiple workers in each VM, isolating them number of URLs, IEWA includes a URL navigation en- from one another using different UIDs and different remote gine, which utilizes IE's extensibility interfaces, such as desktop sessions. IWebBrowser2, to completely automate the browser's nav- On such a setup, one IEWA worker is able to analyze about igation. In addition to pointing the browser to new URLs, 115 typical web sites per hour. Each site's processing time this module also cleans up state such as pop-ups between includes the home page, five random link clicks, and one form submission, as well as overheads introduced by IEWA 's Tables IV, V, and VI present the results of our analysis, three interposition modules. We found that we could execute showing how frequently each feature we analyzed earlier is up to eight parallel workers in one VM, for a throughput of encountered. Next, we organize our findings according to 900 sites per VM, before saturating the CPU. Optimizing our discussion in Section II and discuss their implications this infrastructure for performance was not a goal of this on future browser security policies. paper and is left as future work. B. The interplay of browser resources IV. EXPERIMENTAL RESULTS

1. DOM and Cookies: Cookie usage is extremely pop-

Our analysis in Section II provides an understanding of the ular, and so is their programmatic DOM access via security characteristics of the current access control policies document.cookie, which we found on 81% web sites in browsers. In this section, we complete the other half for reading and 76% of web sites for writing cookie values, of the equilibrium by using the measurement infrastructure respectively. The use of the cookie's domain attribute is presented in Section III to study the prevalence of unsafe also widespread (67% of sites), with about 46% of sites browser features (analyzed in Section II) on a large set of using it to actually change the domain value of the cookie. popular web sites. By presenting both sides, we enable the As a result, the issues described in Section II-C1 cannot browser vendors to make more informed decisions about be solved by simply deprecating the usage of this attribute whether or not to continue supporting a particular unsafe and changing the principal definition of cookies. One pos- feature based on its real-world usage. sible approach to solve the inconsistency issue with cookie handling is to tag the cookie with the origin of the page A. Experimental overview setting the cookie. This information should be passed to the

1. Choosing the sites for analysis: Instead of randomly server to allow the server to differentiate between duplicate

crawling the web and looking for unsafe features, we de- cookies. cided to focus our attention on the "interesting" parts of the Section II-C1 also identified inconsistencies pertaining web that people tend to visit often. Accordingly, to seed our to cookies and HTTP/HTTPS, which we now support analysis, we take the set of 100,000 most popular web sites with measurements. First, 0.07% of sites alarmingly send ranked by Alexa [5], as seen on November 9, 2009, as our secure cookies over HTTP. This effectively tampers with representative data set. The data collection and analysis were the integrity of cookies that may have been intended for completed in the last week of February 2010. HTTPS sessions [10]. Fortunately, it appears that this func-

1. Defining the compatibility cost: We define the cost of tionality can be disallowed with little cost. Surprisingly, a

removing a feature to be the number of Alexa-ranked, top much larger number of sites (5.48%) sent HTTP cookies 100,000 sites that use the feature. over HTTPS. The HTTP cookies cannot be kept confidential We conservatively assume that disallowing a feature will and are accessible to HTTP sessions. Our recommended significantly hinder a site's functionality, whereas it could solution to this problem is that the "secure" flag should be simply cause a visual nuisance. A more detailed analysis on enforced for any cookies passed over an HTTPS connection the effect of policy changes on page behavior is promising even if the web developer fails to set the flag. This would but is left as future work. still enable the HTTPS site to access the cookie for its own

1. High-level results: We obtained our results by ren- functionality and any sharing with the HTTP site should be

dering each of the 100,000 seed links using WebAnalyzer, done explicitly. saving all interposition logs for offline analysis. This way, We found a large number of sites (16.2%) using HttpOnly we were able to obtain data for 89,222 of the 100,000 sites. cookies, which is an encouraging sign — many sites appear There are several reasons why no data was produced for the to be tightening up their cookie usage to better resist XSS rest of sites. First, some sites could not be accessed at the attacks. time of our analysis due to failed DNS lookups, "404 Not 2) Cookies and XMLHttpRequest: Our measurements Found" errors, and other similar access problems. Second, show that the issues arising from undesirable interplay of some sites timed out within our chosen threshold interval of XMLHttpRequest and HttpOnly cookies (Section II-C2) 2 minutes, due to their slow or continuous rendering. We can possibly be eliminated, since very few sites (0.30%) decided to drop any such sites from our analysis. Finally, manipulate cookie headers in XMLHttpRequest responses. some sites did not contain any JavaScript code, and as a 3) DOM and Display: Section II-C3 argued that the result they did not trigger our event filters. Nonetheless, we descendant navigation policy is at conflict with SOP for believe that we have been able to analyze a sufficiently large DOM. We observe iframe navigations on 7.7% of sites and set of sites with a reasonable success ratio, and our data set all of them are child navigation (regardless of the origin). and the scope of measurement is much larger than that used The absence of descendant navigation in the top 100,000 by earlier related studies [24]. sites indicates a potentially very low cost to remove it.

Measurement Criteria (count)	Count	Percentage
document.cookie (read)	5656310	72587 81.36%
document.cookie (write)	2313359	68230 76.47%
document.cookie domain usage (read)	2032522	59631 66.83%
document.cookie domain usage (write)	1226800	41327 46.32%
Secure cookies over HTTP	259 62	0.07%
Non-secure cookies over HTTPS	15589	4893 5.48%
Use of "HttpOnly" cookies	33180	14474 16.22%
Frequency of duplicate cookies	159755	4955 5.55%
Use of XMLHttpRequest	19717	4631 5.2%
Cookie read in response of XMLHttpRequest	1261	265 0.30%
Cross-origin descendant navigation (reading descendant's location)	6043	61 0.07%
Cross-origin descendant navigation (changing descendant's location)	0	0 0.00%
Child navigation (parent navigating direct child)	22572	6874 7.7%
document.domain (read)	1253274	63602 71.29%
document.domain (write)	8640	1693 1.90%
Use of cookies after change of effective domain	295960	1569 1.76%
Use of XMLHttpRequest after change of effective domain	225	87 0.10%
Use of postMessage after change of effective domain	0	0 0.00%
Use of localStorage after change of effective domain	42	10 0.01%
Use of local storage	1227	169 0.19%
Use of session storage	0	0 0.00%
Use of fragment identifier for communication	5192	3386 3.80%
Use of postMessage	6523	845 0.95%
Use of postMessage (with no specified target)	0	0 0.00%
Use of XDomainRequest	527	125 0.14%
Presence of JavaScript within CSS	224266	4508 5.05%

Table IV U SAGE OF VARIOUS

BROWSER FEATURES ON POPULAR WEB SITES (FEBRUARY 2010). ANALYSIS INCLUDES 89,222 SITES .

Sites containing at least one <iframe> 36549 (40.8%) Average number of <iframe>'s per site 3.2
Sites with at least one pair of overlapping frames 5544 (6.2%) Sites with at least one pair of
overlapping cross-origin frames 3786 (4.2%) Sites with at least one pair of transparent overlapping
frames 1616 (1.8%) Sites with at least one pair of transparent overlapping cross-origin frames 1085
(1.2%) Table V SUMMARY OF DISPLAY LAYOUTS OBSERVED FOR THE TOP 100,000 ALEXA WEB
SITES (DECEMBER 2009). 89,483 SITES WERE RENDERED SUCCESSFULLY AND ARE INCLUDED IN
THIS ANALYSIS .

In addition, we have analyzed the visual layouts of all sites at least one overlapping pair of iframes
— but only 29% to determine whether there are dangerous pixel interplays of these overlaps
involved transparent iframes. Most (68%) between windows of different principals (Section II-C3).
Our overlapping scenarios involve different principals. results are summarized in Table V1 . We
found that 41% of The most dangerous situations occur when a transparent sites embed at least
one iframe, and the average number of frame is overlaid on top of a frame belonging to a dif-
iframes embedded on a particular page is 3.2. Overlapping ferent principal (Section II-C3). We
identified 1,085 sites iframes appear to be common — 6.2% of sites contained (1.2%) that
contained at least one pair of transparent, cross- origin overlapping iframes. We observed that
most of these 1 Our display analysis was performed in December 2009, separately from overlaps
involved domains serving ad banners, so the main script engine and network analysis that we
performed in February 2010, causing a slight difference in the number of successfully rendered
sites in site functionality might remain unaffected if the dangerous Tables V and IV. transparency is
disallowed. Total instances Unique sites Measurement Criteria (count) Count Percentage Setting
top-level window's location 55759 2851 3.20% Change focus of window 5221 2314 2.59% Reading
color of hyperlinks 82587 1560 1.75% Accessing browser's history 1910 721 0.81% Use of
defaultStatus (write) 1576 241 0.27% Reading user's Geolocation 251 149 0.17% Use of resizeTo
339 134 0.15% Use of defaultStatus (read) 528 108 0.12% Use of moveTo 258 100 0.11% Close a
window 130 86 0.10% Access to user's clipboard 24 17 0.02% Blur a window 54 13 0.01% Use of
resizeBy 13 8 0.01% Use of moveBy 4 1 0.00% Use of outerWidth 2 1 0.00% Use of outerHeight 4 1
0.00% Table VI P REVALENCE OF RESOURCES BELONGING TO THE USER PRINCIPAL ON POPULAR
WEB SITES . ANALYSIS INCLUDES 89,222 SITES .

Summary. We found that interplays between DOM and document.domain completely carries a
substantial cookies have a high compatibility impact, while removing cost (1.9% of sites), browsers
can eliminate its impact on the interplays between cookies and XMLHttpRequest would
XMLHttpRequest, local storage, and postMessage at a much affect only 0.30% of sites. For
interplays related to display, lower cost (0.19% of sites total). On the flip side, browser we found
that descendant navigation can be disallowed vendors have to make a much tougher choice
(affecting with no cost, while disallowing overlaps between transparent 1.76% of sites) to prevent
effective UID inconsistencies cross-origin frames would affect 1.2% of sites. pertaining to cookies.
D. Resources belonging to the user principal C. Changing effective Principal ID Table VI shows the
results of our analysis for the cost of In Section II-D, we showed that document.domain is
protecting user-owned resources discussed in Section II-E. an unsafe and undesirable part of
today's web, as observed The cost of tightening access control for user resources by others as well

[9]. Unfortunately, we found its usage on appears to be low with the exceptions of link-color access the web to be significant: 1.9% of sites change their effective (1.8%), the focus-changing functions (2.6%), and setting domain via document.domain. top-level window location (3.2%). We mentioned certain features which become incoherent Interestingly, 149 sites (0.17%) already use the new when combined with document.domain. Cookies are Geolocation primitives [12]. This number seems low enough accessed by about 1.76% of the sites after a change in for browsers to take actions to tighten its access control. effective domain, making it difficult to enforce a unified Overall, we found that 12 of the 16 user-principal APIs effective domain for cookie access (Section II-D1). Only we examined can be removed while collectively affecting 0.08% of sites use XMLHttpRequest after an effective UID only 0.80% of unique sites. change (Section II-D2), so it appears possible to make XMLHttpRequest respect effective domain with little cost. E. Other noteworthy measurements The same holds true for postMessage — we found no We measured prevalence of some primitives for cross- sites using postMessage after an effective UID change. The frame and cross-window communication, which are critical new local storage abstractions are not widespread — only for cross-principal security. Fragment identifier messaging 0.19% of the sites were using localStorage (0.01% is most popular, being found at 3.8% of sites. A non- after an effective domain change), and no sites were us- negligible number (0.95%) of sites have already adopted ing sessionStorage — so we anticipate that origin- postMessage, and all sites use its newer definition that changing weaknesses that we outlined in Section II-D4 can requires specifying the target window [13]. Another safer be removed with little compatibility cost. alternative for cross-domain communication, XDomainRe- Summary. Overall, while disallowing quest, is also being slowly adopted (0.14%). 100 defaultStatus (read) document.domain mutation defaultStatus (write) PostMessage moveTo Frame Identifier Messaging resizeBy 80 resizeTo location blur focus 60 close geolocation CDF

clipboard color 40 history

1 10 100 1000 10000 100000 Popularity Ranking (log) 20 Figure 7. A CDF for prevalence of user-owned resources according to the ranking of sites that use them. 0 0 20000 40000 60000 80000 100000 Sites ranked by popularity

Figure 6. A CDF for prevalence of cross-frame communication mecha- G. Methodology validation using user-driven analysis nisms according to the ranking of sites that use them. In the previous sections, we examined sites by visiting their home pages and relying on WebAnalyzer’s heuristics (see Section III) to simulate a few basic user actions to Using JavaScript within CSS has long been considered invoke additional functionality that may be hidden behind dangerous [9]. We found this pattern in use on about 5% of “splash” home pages. However, our methodology may miss the sites. site functionality that requires user login forms (e.g., on Facebook), other more sophisticated user event handlers (e.g., mouse movements), or following many links away F. Correlating unsafe features and site popularity from the home page. In general, it is very difficult, if not impossible, to simulate user actions that open access to Next, we consider how the popularity of sites correlates representative features of an arbitrary site. with prevalence of unsafe features. A policy is more costly To evaluate the limitations of our heuristics-driven ap- to correct if it is used by very highly ranked sites, since more proach, we conducted a user-driven examination of the top people would visit them and encounter broken functionality. 100 Alexa sites. To do this, one of the authors manually Fortunately, we found that most features do not exhibit visited these sites with IEWA and

used his best judgement a significant popularity bias, behaving uniformly with no to invoke the site's representative functionality. For example, regard to a site's popularity. Nevertheless, we found some for analyzing Facebook, the author logged into his Facebook exceptions. Figure 6 shows a CDF of the usage of various account, browsed through several profiles, and invoked sev-

mechanisms that could be used for cross-frame communica- eral applications such as photo viewing or messaging. tion according to the sites' ranking. Interestingly, fragment We then compared the results obtained through this man- identifier messaging has little dependence on popularity, ual analysis to those obtained using WebAnalyzer for the document.domain tends to be used more by higher- same sites. Table VII summarizes the results of our compar- ranked sites, and postMessage is found more on lower- ison. We observe that the numbers of sites using a particular ranked sites, with very little use in the top 2000 sites. This feature are mostly comparable, providing confidence that our went against our hypothesis that higher-ranked, high-profile heuristic-driven navigation engine in WebAnalyzer works sites would likely be written using the latest and safest web well in practice. Some features have higher prevalence standards. A possible explanation could be that the top sites with the user-driven analysis, as expected, but there are are motivated to use features compatible with the largest only a couple of outliers. For example, Geolocation was number of browsers and client platforms. found on nine sites, all found on multilingual versions of As another example, Figure 7 diagrams the prevalence of maps.google.com. In manual analysis, the user invoked resources belonging to the user principal according to the maps on each of the nine versions of the Google site, ranking of the sites that use them (a dot is displayed for where WebAnalyzer randomly picked and followed the link every site using a particular feature). Some features, such to Maps on three of these sites. On the other hand, on as resizeBy or clipboard access, are only found on very several occasions, WebAnalyzer also found features that low-ranked sites and are thus good candidates to remove were missed by manual analysis, as can be seen in higher with little impact. Only a handful of features appear in the prevalence for features like reading document.domain. top 100 sites, where compatibility cost is very high for any This can happen when WebAnalyzer navigates to a link site. that the user did not examine as part of representative Number of sites Measurement Criteria WebAnalyzer Manual

document.cookie (read)	93	86
document.cookie (write)	86	76
document.cookie domain usage (read)	78	70
document.cookie domain usage (write)	59	59
Secure cookies over HTTP	0	2
Non-secure cookies over HTTPS	11	8
Use of "HttpOnly" cookies	27	30
Frequency of duplicate cookies	17	8
Use of XMLHttpRequest	32	28
Cookie read in response of XMLHttpRequest	0	0
Cross-origin descendant-navigation (reading descendant's location)	0	0
Cross-origin descendant-navigation (changing descendant's location)	0	0
Child navigation (parent navigating direct child)	1	2
document.domain (read)	78	59
document.domain (write)	18	19
Use of cookies after change of effective domain	18	19
Use of XMLHttpRequest after change of effective domain	4	2
Use of localStorage after change of effective domain	2	1
Use of session storage	0	0
Use of local storage	4	3
Use of fragment identifier for communication	0	1
Use of postMessage	1	1
Use of XDomainRequest	1	2
Presence of JavaScript within CSS	16	27
Setting top-level window's location	1	2
Change focus of window	2	2
Reading user's Geolocation	3	9

Table VII C OMPARISON OF USER - DRIVEN ANALYSIS VS . W EBA NALYZER FOR THE TOP 100 A LEXA SITES . F EATURES NOT SHOWN HERE WERE USED BY ZERO SITES FOR BOTH USER - DRIVEN AND W EBA NALYZER STUDIES .

functionality on a given site. Overall, we felt our heuristics- new one identified 32 such pages (see Table VII). One of driven approach achieved good coverage, though larger-scale the reasons is that

many search sites use XMLHttpRequest user-driven measurements would still be very valuable in to auto-complete the search string as users type it; our old complementing WebAnalyzer measurements. system did not trigger this behavior, whereas our new system triggered it when auto-filling the search textbox. Many other V. D DISCUSSION AND LIMITATIONS features showed a similarly dramatic jump in prevalence. Benefits of heuristics-driven automated crawling. In Limits of automated crawler-based measurements. AI- our original design, WebAnalyzer visited only the top- though we believe that our resulting measurements provide a level page of each site we studied. We quickly realized good representation of the use of browser features on popular that this analysis failed for sites that hide much of their web sites, it is likely that we missed certain features because functionality behind “splash” home pages. This became most the code path to invoke them was not triggered in our apparent when studying the original results for Table VII. We analysis. For example, sites like Facebook or banks require a observed that for many sites, clicking on a link or filling out user to sign in, game sites require particular mouse gestures a search form on the home page would expose a noticeably to invoke certain behavior, and numerous sites require appro- larger (though still not complete) set of functionality. Thus, priate text (such as stock symbols or user’s personal data) to we augmented WebAnalyzer with simple heuristics that be entered into forms. Even if we could solve some of these imitate this user behavior (see Section III). problems, for example by enumerating all events registered As an example, our original system saw XMLHttpRequest on a page or using a database of dummy usernames and calls on only 13 pages of the top 100 pages, whereas the passwords [27], automatically invoking certain features, such as buying products on shopping sites, is inappropriate. This of these other segments of the web as important future work. ultimately limits our ability to explore all features invoked on today’s web. VI. R ELATED W ORK We also did not try to exhaustively crawl each site. Even in We are not the first to find and analyze flaws in browser our user-driven analysis (Section IV-G), we did not attempt security policies. Previous work has looked at weaknesses in to enumerate and invoke all gadgets on every page of each cross-frame communication mechanisms [13], frame naviga- site. Thus, the results we collect for a particular site cannot tion policies [3], [13], client-side browser state [21], cookie be used as a list of all features the site might have. Our aim path protection [28], protection among documents within was to favor breadth over depth and obtain good coverage same origin [2], display protection [3], and other issues. for the representative features of 100,000 sites we tested. Zalewski [9] documents the security design in browsers While our infrastructure could also be used for exhaustively including some loopholes. This work complements these crawling each site, we would need to dramatically scale up efforts by identifying incoherencies in browser’s access our current infrastructure to cover a comparable number of control policies. To our knowledge, this is the first principal- sites, and we leave this as future work. driven analysis on browsers’ access control policies. Picking the right browser. Some sites check the client’s DOM access checker [22] is a tool designed to au- browser version (using the user-agent header) before de- tomatically validate numerous aspects of domain security ciding to invoke a particular code path. Although not a policy enforcement (cross-domain DOM access, JavaScript base requirement, we developed WebAnalyzer with IE as cookies, XMLHttpRequest calls, event and transition han- the underlying browser. This could prevent code invocations dling) to detect common security attacks or information that are intended for non-IE browsers, thereby leading to disclosure vectors. Browserscope [29] is a community-driven missed features. For example, XMLHttpRequest2 [15] is project for tracking browser functionality. Its security test

currently not supported by IE, and it would be missed suite [23] checks whether new browser security features by WebAnalyzer if the site invokes it only after verifying are implemented by a browser. In our analysis of access browser support. control policies, we uncovered incoherencies by examining A related problem is fallback code that invokes an alter- the interplay between resources, runtime identity changes, native implementation of a feature that a browser doesn't and the user principal's resource access control. This focus support. For example, a site could first check whether the and methodology differ from this previous or ongoing work, browser supports postMessage for cross-frame commu- and our analysis not only touches on DOM, but also on the nication, and fall back on fragment identifier messaging if it HTTP network layer and display. Nevertheless, we plan to does not. Because we use IE 8, we will log that this site uses contribute our test programs to one of these test suites. postMessage, but older browsers would utilize fragment Compared to previous work, a unique aspect of this identifier messaging. work is our extensive evaluation of the cost of removing The compatibility cost of features invoked in browser-unsafe policies from the current web by actively crawling dependent code paths depends not only on the number and executing web content. Yue et al. [24] also used a of web sites using a feature, but also on the number of crawling-based, execution-based approach to measure the visitors utilizing a particular browser that relies on such prevalence of unsafe JavaScript features on 6805 popular code. Evaluating the second part of this cost is orthogonal web sites. They used a JavaScript interposition technique to our goals in this paper: rather than exploring prevalence that is similar to IEWA 's script engine proxy, but they of features on web sites, it asks how many of a web site's lack IEWA 's network and display interposition capabilities, clients rely on a particular browser. Web server operators limiting the policies they can monitor. As well, we present can easily answer this question by profiling "user-agent" results from a significantly larger dataset. strings in incoming HTTP requests. As future work, we Our active crawling infrastructure builds on previous can integrate other browsers into WebAnalyzer, or we can efforts that have analyzed safety of web pages by rendering modify IEWA to render a site with a set of user-agent strings them in real browsers running within virtual machines [30]– representing other browsers; this would capture a more [34]. We extend these frameworks with additional browser complete set of the site's code. interposition support to monitor unsafe browser security Studying other web segments. Our focus on the top policies. 100,000 sites represents a particular segment of the web with a good balance of the very top sites and some of the VII. C ONCLUSIONS less popular "tail". However, this still covers only a tiny In this paper, we have examined the current state of fraction of the billions of pages on today's web. In addition, browser access control policies and analyzed the incoheren- our analysis excluded intranet sites, which are hidden from cies that arise when browsers mishandle their principals traditional crawlers, and which can influence backwards by (1) inconsistently labeling resources with principal IDs, compatibility decisions for a browser. We leave exploration (2) inappropriately handling principal identity changes via document.domain, and (3) neglecting access control for [16] "Mitigating Cross-site Scripting With HTTP-only Cookies," certain resources belonging to the user principal. In addition <http://msdn2.microsoft.com/en-us/library/ms533046.aspx>. to pointing out these incoherencies, we have developed a Accessed on Nov. 14, 2009. [17] "HttpOnly," <http://www.owasp.org/index.php/HTTPOnly>. web compatibility analysis infrastructure and measured the Accessed on Nov. 14, 2009. cost of removing many unsafe policies we identified for a [18] "Mozilla Foundation Security Advisory 2009-05: large set of popular web sites. Overall, this work contributes XMLHttpRequest allows reading HTTPOnly

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